

ES51 Turf Wars Competition Final Report

Team 3: Jack in the Bot

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Abstract

This project aimed to apply mechanical engineering design and manufacturing principles by developing a remote controlled robot that could complete the objectives of the "Turf Wars" competition and be produced by a combination of all the manufacturing techniques learned throughout the course. The primary objective of the Turf Wars robot was to efficiently collect, transport, and deposit game objects while maintaining mobility across ramps and within strict dimensional and actuator constraints. Our team approached this challenge by prioritizing mechanical simplicity, robustness, and multifunctional components to maximize performance within a limited motor budget. The final design consisted of a continuous rotational intake and storage mechanism integrated with a compact drivetrain, which allowed the robot to move and collect multiple objects simultaneously, tolerate misalignment, and operate with high efficiency. Overall, the robot performed reliably during testing and competition, demonstrating strong scoring capability and stable mobility.

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1 Introduction

ES51: Computer-Aided Machine Design is Harvard’s introductory Mechanical Engineering course. The final project, known as “Turf Wars,” is a remote-controlled robot competition that brings together the concepts developed throughout the semester. In this project, teams design and manufacture robots to compete in a controlled arena and complete task-based objectives.

The Turf Wars competition requires robots to navigate obstacles, manipulate a variety of objects, and maximize scoring within a limited time. All designs must satisfy strict constraints related to size, materials, and manufacturing processes. Robots are developed throughout the semester, primarily during weekly lab sessions, where teams have access to machining and fabrication facilities. Each team is also provided with a standard materials kit, including acrylic sheets and components derived from cordless screwdrivers. These resources must be used to create a robot that balances performance, durability, and manufacturability.

Within this context, the design challenge extends beyond basic functionality. Effective solutions require the integration of mobility, object manipulation, and strategic operation into a cohesive system that performs reliably under competition conditions.

Team “Jack in the Bot” addressed this challenge by prioritizing simplicity, robustness, and scoring efficiency. Rather than focusing on precise, single-object manipulation, the design emphasizes bulk collection and delayed scoring to reduce cycle time and increase point throughput. This approach led to the development of a compact, integrated mechanism capable of performing intake, storage, and scoring through a single rotational motion.

This report presents the design rationale, engineering considerations, manufacturing process, and iterative development that resulted in the final robot, the “Sooper Scooper 5000.”

2 Concept Development

2.1 Constraints

The design of the robot is limited by several key constraints imposed by the competition rules and fabrication environment.

- **Size:** The robot must fit within an 11 in $\times$ 11 in $\times$ 11 in volume at the start of the match. This limits the initial footprint and constrains mechanism geometry.
- **Terrain and mobility:** The robot must traverse ramps of up to 30 degrees and operate on uneven turf surfaces. This requires sufficient traction, careful weight distribution, and adequate torque.
- **Scoring geometry:** Objects must be deposited into an inventory zone with a height of approximately 5 inches. This creates a requirement for a lifting or elevation mechanism within a compact design.
- **Materials and manufacturing:** The robot is fabricated using available shop processes, including laser cutting, 3D printing, and manual machining, and relies on a limited set of provided materials. This limits achievable tolerances and geometric complexity.
- **Time and prototyping:** The project timeline is limited by the pace and overall length of the semester. This limits the number of possible design iterations and testing cycles, which reduces opportunities for refining complex mechanisms.

These constraints were a strong influence the team's design approach. The size and scoring requirements encouraged a compact mechanism that integrates lifting and storage. Terrain constraints required a stable drivetrain with sufficient torque and traction, reinforcing a low center of mass. Manufacturing and time limitations favored simple, robust solutions over more complex alternatives.

2.2 Criteria

To evaluate potential design concepts, a set of performance-driven criteria is established based on the competition objectives and constraints, as well as the manufacturing limitations. These criteria focus on the robot’s ability to efficiently collect, transport, and score objects while maintaining reliability and simplicity.

- **Scoring efficiency:** Ability to collect and deposit multiple objects quickly to maximize points within the time limit.
- **Mobility and terrain traversal:** Capability to navigate ramps up to 30 degrees and operate effectively on uneven turf surfaces.
- **Mechanical simplicity:** Minimization of moving parts and system complexity to improve reliability and ease of fabrication.
- **Manufacturability:** Feasibility of fabrication using available tools, materials, and time constraints.
- **Robustness and durability:** Ability to withstand collisions, repeated loading, and multi-round competition without (or with minimal, fixable) failure.

2.3 Design Specifications

The design constraints and criteria defined in Sections 3.1 and 3.2 were translated into a set of quantitative technical specifications that served as concrete engineering targets throughout the design process. These specifications [Table 1] functioned as a practical design checklist during brainstorming and iteration, ensuring that each proposed concept could be evaluated against measurable performance requirements. Rather than representing a final summary of the design, they were actively used to guide decision-making and maintain consistency across all iterations.

Specification	Target Value	Rationale / Source
Geometry and Dimensions		
Intake throat width	≥ 4.67 in	Must span one full inventory bin cell (4.67 in TYP per Figure 3)
Throat height clearance	≥ 1.6 in	Ping-pong ball diameter ≈ 1.57 in; add 0.03 in clearance
Scoop lip angle	$10^\circ\text{--}20^\circ$ from ground	Must slide under flat pucks without catching on floor surface
Scoop tip ground clearance	≤ 0.1 in	Pucks are low-profile; tip must reach under them
Retracted envelope	$\leq 11 \times 11 \times 11$ in	Box of Justice starting constraint (rules section 1.6.1)
Delivery height	≥ 5.0 in	Inventory Zone opening height (rules section 1.2)
Forces & Torque — Kinetics		
Object load (worst case)	≈ 2 N (200 g puck)	Heaviest single game object; conservative estimate
Min. intake shaft torque (flat)	$\tau \geq 0.10$ N $\cdot$ m	$\tau = F \times r = 2 \text{ N} \times 0.025 \text{ m} \rightarrow 0.05 \text{ N} \cdot \text{m} \times 2 \times \text{safety factor}$
Min. intake shaft torque (ramp)	$\tau \geq 0.11$ N $\cdot$ m	$+ \sin(15^\circ) \times 0.2 \text{ kg} \times 9.8 \approx 0.51 \text{ N}$ added for tennis ball on 15° ramp
Lateral arm deflection limit	≤ 2 mm permanent set at 5 N lateral load	Survive repeated arena collisions without plastic deformation
Speed & Cycle Time — Kinematics		
Intake surface speed	≥ 0.3 m/s	Must ingest objects without robot needing to stop

Specification	Target Value	Rationale / Source
Full intake cycle time	≤ 2 s (contact $\rightarrow$ secured)	4-minute round; must allow multiple pickups
Mass & Material — Weight		
Intake assembly mass budget	≤ 200 g	Total robot limit 2.27 kg; reserve majority for drivetrain + chassis
Structural plate material	Acrylic ≥ 6 mm or Al 6061 ≥ 3 mm	Lab-approved materials; acrylic ($\rho = 1.18$ g/cm ³) preferred for weight
3D printed parts (PLA)	$\rho = 1.24$ g/cm ³ ; within \$70 budget	Low-load brackets only; track spend per rules §1.8.9
Power & Electrical — Power		
Intake actuator current draw	$\leq 2\text{--}3$ A continuous	Total motor budget ≤ 9 A; drivetrain uses $\approx 6\text{--}7$ A (rules section 1.9.3)
Servo stall torque (if servo used)	≥ 0.10 N·m at operating voltage	Must meet shaft torque requirement above at actual supply voltage

Table 1: Design Specifications

2.4 Design Iterations

Over the course of the project, numerous design concepts were explored, ranging from early sketches to physical prototypes. This section presents the most mechanically viable candidates that emerged from that process and were advanced to formal evaluation.

2.4.1 Design 1: Overhead Scoop

The first design concept centered on a large rear-mounted scoop that rotates upward and over the robot’s body to deposit collected objects into the inventory bins. In its open position, the scoop rests near ground level at the front of the robot, allowing it to slide beneath game objects. Once loaded, a motor-driven rotation arcs the scoop overhead, releasing the objects into the scoring zone

behind the robot.

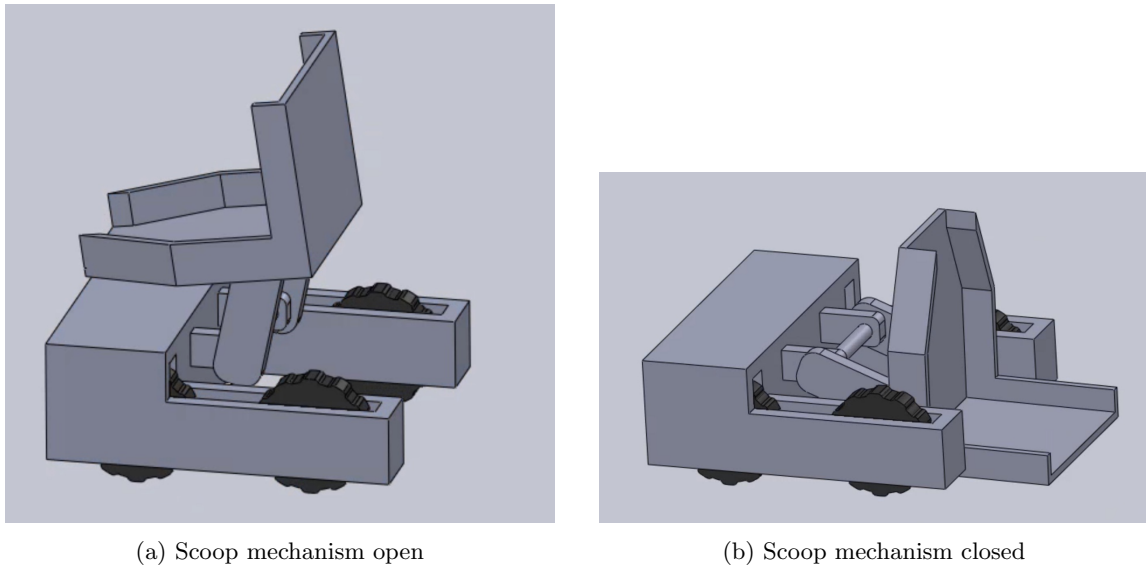


Figure 1: Design 1 "Perspective view" screen captures of SolidWorks CAD model

Advantages:

- The large scoop geometry allows for bulk collection of multiple objects in a single pass, reducing the number of trips required to the scoring zone.
- The mechanism requires relatively few components, keeping the overall part count low.

Disadvantages:

- The overhead rotation significantly raises the center of mass when the scoop is lifted, which makes the robot more susceptible to tipping, particularly when traversing the 30° ramps required in the competition.
- Actuating the scoop through its full range of motion requires relatively high torque. This in turn requires a dedicated servo that would consume a large portion of the available motor budget.

- When raised, the scoop also occupies space above the main chassis, which interferes with packaging for key components such as the battery and drivetrain motors. This creates internal layout constraints that reduces overall design flexibility.
- The mechanism relies on the robot positioning itself against a wall or boundary to effectively collect objects, which limits maneuverability and reduces operational flexibility during gameplay.

2.4.2 Design 2: Horizontal Scoop Intake

The second design concept used a crank-driven sweeper mechanism mounted horizontally at the front of the robot. A motor-driven circular cam actuated a linkage arm connected to a flat sweeper plate positioned close to the ground. As the cam rotates, the linkage converts rotational motion into a reciprocating sweeping action that pushed objects laterally into a collection zone. A cardboard prototype was constructed to validate the kinematic behavior before committing to full fabrication.

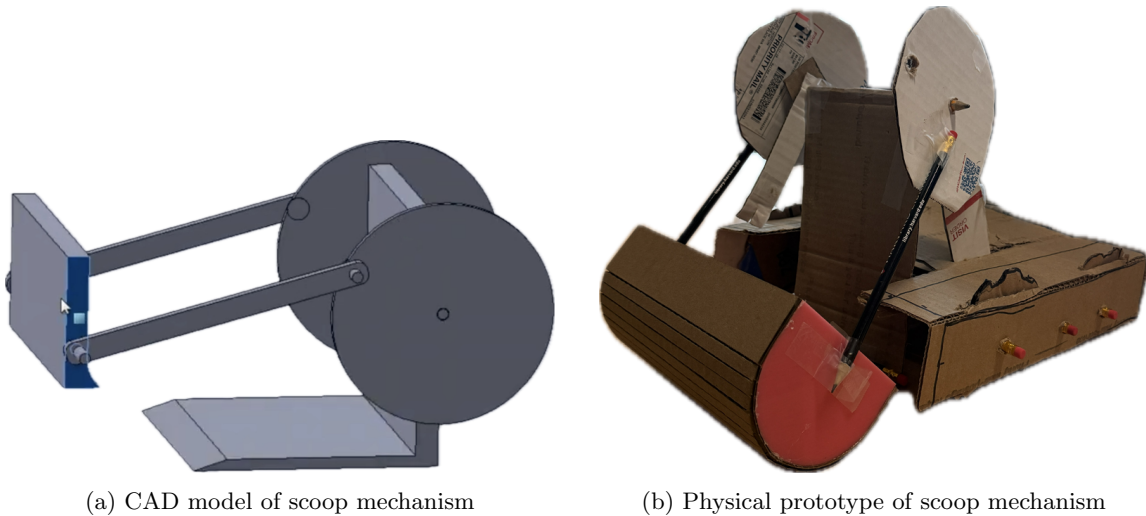


Figure 2: Scoop mechanism CAD and prototype

Advantages:

- The wide sweeper plate provided a large effective capture area, enabling the robot to collect

multiple objects in a single pass without requiring precise alignment. This improved robustness during operation, particularly under the time-constrained and dynamic field conditions.

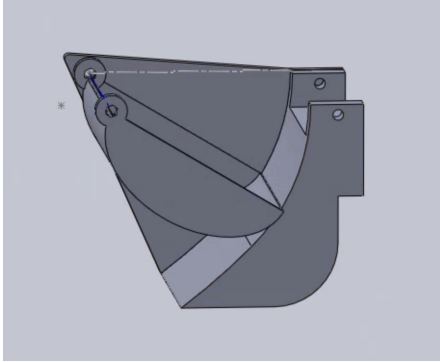
- The mechanism was also geometrically straightforward, relying primarily on planar linkage motion. This made it relatively easy to prototype and allowed for early-stage validation of the core kinematic concept.

Disadvantages:

- The crank-sweeper linkage requires at least two additional degrees of freedom beyond the drivetrain, exceeding the available motor budget. Since both drive motors were already allocated, there was no remaining actuator capacity to independently drive this mechanism.
- The linkage introduces sensitivity to geometric tolerances. Small deviations in linkage length or pivot placement result in significant variations in the sweep path, reducing reliability in consistent object capture.
- The reciprocating motion limits operational speed compared to continuous rotational intake systems, reducing overall output rate under competition time constraints.

2.4.3 Design 3: Rotating Scoop (Sooper Scooper 5000)

The rotating scoop mechanism consists of a ramp-and-scoop assembly driven by a single motor connected to a central shaft. As the shaft rotates, the semicircular scoop sweeps along the ground, collecting objects via the ramp and carrying them upward. At 180° of rotation, the open face of the scoop faces inward, effectively trapping the collected items against the robot body, allowing the robot to drive freely while objects remain secured. Completing the full 360° rotation returns the scoop to its original position, releasing the items into the designated scoring zone. This single continuous motion therefore achieves intake, storage, and deposit in one seamless action, requiring no additional actuators or mechanisms.



(a) CAD model of scoop mechanism



(b) Final robot on competition field

Figure 3: Final robot and scoop CAD design

Advantages:

- The mechanism integrates intake, storage, and deposit into a single continuous rotational system driven by one motor and shaft. This reduces mechanical complexity, eliminates the need for additional actuators, and simplifies system coordination while improving overall reliability through fewer potential failure points.
- The ramp element design enables efficient bulk pushing, allowing multiple objects to be shoved into "trash areas" without requiring precise alignment. Its passive guiding geometry improves robustness by tolerating misalignment and directing objects into the intake without strict positioning requirements.
- The compact configuration maintains a low center of gravity, enhancing stability during operation and enabling reliable traversal of inclined surfaces up to 30° without compromising control or balance.
- The overall simplicity of the mechanism also supports rapid fabrication and assembly within constrained resources and timeframes, making it well-suited for iterative prototyping and competition deployment.

Disadvantages:

- Deposit location is constrained by the robot’s orientation, requiring the driver to align the robot correctly before completing the rotation.
- The semicircular geometry limits scoop volume, meaning very large objects may not be fully secured during the storage phase.
- Bulk pick up may reduce time-efficiency given that items must be dropped at every full rotation. Items either have to be shoved closer together with the ramp element, or dropped and re-picked up along other items.

2.5 Final selection

2.5.1 Pugh Matrix

A passive scoop, in which the robot simply pushes objects without any active mechanism, was used as the neutral baseline for comparison purposes.

Criterion	Baseline Passive Scoop	Design 1: Overhead Scoop	Design 2: Horizontal Scoop	Design 3: Rotating Scoop
Scoring efficiency	0	+	+	++
Mobility / traversal	0	--	-	+
Mechanical simplicity	0	-	--	++
Manufacturability	0	-	--	++
Robustness	0	-	-	+
Total	0	-4	-5	+7

Table 2: Pugh chart comparing design concepts relative to a passive scoop baseline.

2.5.2 Selection Rationale

After evaluating all alternative concepts against the established criteria, the rotating scoop mechanism was selected as the final design solution. The primary driver behind this decision was the team’s design philosophy that mechanical simplicity is a prerequisite for reliability and operational

efficiency. A system with fewer moving parts has fewer points of failure, is easier to manufacture within the constraints of the available fabrication resources, and is more predictable under competition conditions. The rotating scoop embodies this philosophy by consolidating intake, storage, and deposit into a single continuous 360° rotation driven by one motor and one shaft, eliminating the need for additional actuators or complex linkages that characterized the alternative designs. Beyond simplicity, the mechanism demonstrated clear advantages in scoring efficiency: its bulk collection capability and single-motion operation significantly reduce cycle time, allowing the robot to collect and deposit multiple objects rapidly within the four-minute match window. The low-profile geometry of the assembly also preserves a low center of gravity, directly supporting the mobility requirements imposed by the 30° ramp constraint. Taken together, these qualities made the rotating scoop the design that best satisfied both the technical constraints and the performance criteria established at the outset of the project.

3 Final System Design and Performance Analysis

To verify that the final design satisfies all technical specifications, a series of engineering analyses were conducted spanning the drivetrain, intake mechanism, motor performance, and structural integrity. In this section, the key results of these analyses are presented alongside the corresponding subsystem designs, directly linking quantitative predictions to observed performance. Detailed calculations, including full derivations, free body diagrams, and SolidWorks FEA results, are provided in Appendix A.

3.1 System Overview

The final robot, the *Soooper Scooper 5000*, consists of three primary subsystems: a two-wheel differential drivetrain, a rotating scoop intake mechanism, and a compact structural chassis that integrates all components within the required dimensional constraints.

The drivetrain provides mobility and ramp-climbing capability through a geared DC motor system. The rotating scoop mechanism, driven by a single motor, performs intake, storage, and scoring

through a continuous 360° rotation. The chassis supports these subsystems while maintaining a low center of mass and ensuring structural integrity under operational and impact loading. Full technical drawings are provided in Appendix D.

3.2 Drivetrain

The drivetrain was designed to enable the robot to climb a 30° ramp while maintaining a target velocity of approximately 0.4 m/s. This requirement imposed constraints on both torque and output speed.

From the analysis, the force required to climb the ramp was calculated as 13.24 N (see [Figure 6] in Appendix), resulting in a required torque of 0.311 N·m per gearbox (the corresponding free body diagram is provided in [Figure 6]). A gear ratio of $\frac{2}{3}$ was selected to match the required output speed while maximizing torque delivery at the wheel.

Parameter	Value
Required torque (per gearbox)	0.311 N·m
Available torque (at wheel)	1.3875 N·m
Gear ratio	$\frac{2}{3}$
Required speed	81.27 RPM

Table 3: Key drivetrain analysis results. Full calculations provided in Appendix A.1.

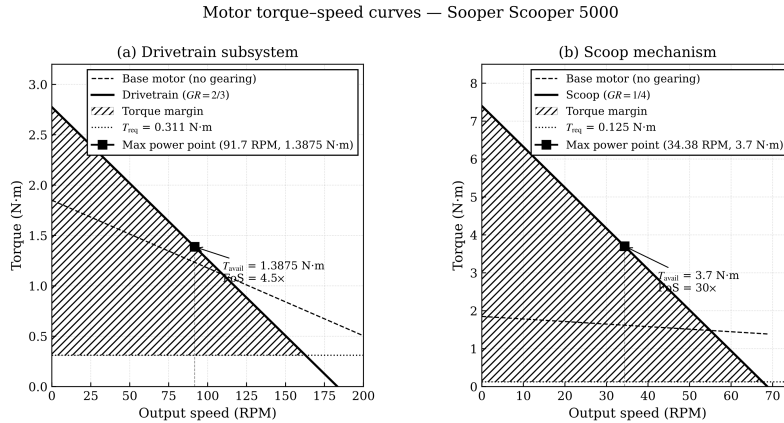


Figure 4: Motor Torque vs. Speed Curves

The available torque exceeds the required torque by more than a factor of four, indicating a strong margin for losses due to friction and uneven terrain.

In practice, the drivetrain performed reliably under competition conditions. The robot consistently climbed ramps without stalling and maintained stability due to its low center of mass (The motor torque–speed characteristics are shown in [Figure 4]). No significant wheel slip or tipping was observed. While the drivetrain is somewhat overdesigned in terms of torque capacity, this contributed to robust and predictable performance.

3.3 Scoop Mechanism

The rotating scoop mechanism was designed to integrate intake, storage, and scoring into a single continuous motion, minimizing mechanical complexity and actuator requirements.

The torque required to lift the heaviest expected object was calculated as 0.125 N·m (the free body diagram for the scoop mechanism is provided in [Figure 5]). Using a gear ratio of $\frac{1}{4}$, the available output torque is 3.7 N·m, providing a substantial safety margin.

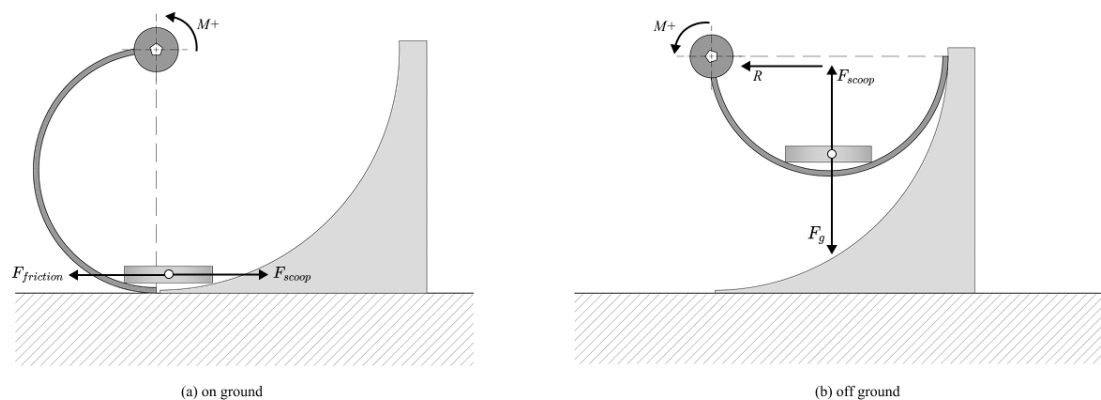


Figure 5: Scooping Mechanism Free Body Diagram

This large torque margin ensures that the mechanism does not stall, even under worst-case loading conditions. The selected gear ratio also limits the rotational speed to a controllable range.

During testing and competition, the mechanism successfully collected multiple objects in a single

Parameter	Value
Required torque	0.125 N·m
Available torque	3.7 N·m
Gear ratio	$\frac{1}{4}$
Max output speed	70 RPM

Table 4: Key scoop mechanism analysis results. Full calculations provided in Appendix A.2.

pass, particularly when objects were clustered. The ramp-assisted geometry reduced the need for precise alignment, improving operational efficiency.

However, the fixed scoop geometry limited the range of object sizes that could be reliably captured at once, and objects could occasionally get lost during transport. Additionally, because objects are released only after a full rotation, scoring required careful alignment of the robot, introducing a dependency on driver control.

Overall, the mechanism achieved its goal of enabling efficient bulk collection while maintaining a simple and reliable design.

3.4 Structural Integrity and Reliability

Structural components were designed to withstand both operational loads and collisions encountered during competition. The most critical structural element analyzed was the aluminum hex shaft driving the scoop. The cross-sectional geometry of the shaft is shown in [Figure 8].

Parameter	Value
Maximum shear stress	4.746 MPa
Material yield strength	207 MPa
Factor of Safety	43.62

Table 5: Structural analysis summary. Full derivation provided in Appendix A.3.1.

The resulting factor of safety indicates that the shaft operates far below its failure limit. While this represents a conservative design, it ensures robustness under uncertain loading conditions.

In practice, the robot experienced repeated collisions and loading cycles without any observable structural failure or permanent deformation. This confirms the reliability of the structural design.

3.5 Integrated System Performance

When evaluated as a complete system, the robot demonstrated strong performance in terms of scoring efficiency and operational reliability. The combination of a high-torque drivetrain and continuous-motion intake enabled multiple collection and scoring cycles within the competition time limit.

The bulk intake strategy proved effective in reducing cycle time, as multiple objects could be collected without requiring precise positioning. However, system performance was influenced by the interaction between subsystems. While intake was efficient, the fixed rotational cycle of the scoop limited flexibility in scoring, occasionally resulting in suboptimal object placement.

Performance also varied with object distribution and size. The system performed best when objects were clustered and within the size range accommodated by the scoop geometry.

Overall, the integrated design achieved a strong balance between simplicity, robustness, and efficiency, though at the expense of precision and control.

3.6 Evaluation Against Criteria and Constraints

The final design satisfied all major constraints established at the outset of the project. The robot remained within the required dimensional envelope, successfully traversed ramps, and achieved the necessary scoring height. All components were fabricated using available manufacturing processes within the project timeline.

In terms of performance criteria, the robot excelled in scoring efficiency due to its bulk collection capability. Mobility was strong, supported by a drivetrain that exceeded torque requirements and maintained stability under load. Mechanical simplicity was a defining strength, as the system relied on a single-actuator mechanism for object manipulation. Manufacturability was also high, with the design using accessible fabrication methods and straightforward geometries.

Notably, during the second round of competition, the scooping mechanism was not activated after the team shared its electronics with another group. The robot therefore operated using only the front ramp as a passive pushing element, limiting scoring to turf and debris zones that did not

require lifting. Despite this constraint, this round resulted in the highest point total achieved by the robot. While external factors such as opponent performance may have influenced the outcome, this result highlights the effectiveness of the system’s overall integration. In particular, the ability to operate efficiently in both active (elevated scoring) and passive (ground-level clearing) modes allowed the robot to remain competitive even when a primary subsystem was unavailable.

The primary limitation was reduced precision in scoring, due to the reliance on continuous rotation and passive object retention.

3.7 Advantages, Limitations, and Future Improvements

A key strength of the design is the integration of intake, storage, and scoring into a single mechanism. This significantly reduces mechanical complexity while maintaining high scoring throughput. The simplicity of the system also enabled efficient fabrication and iteration within the project timeline.

However, this simplicity introduces limitations. The lack of independent control over intake and release reduces precision, and the fixed geometry of the scoop limits the range of objects that can be reliably handled.

Future improvements would focus on addressing these limitations while preserving the advantages of the design. One potential modification is the addition of passive retention features to improve object containment during transport. Another improvement would involve increasing the effective volume of the scoop without significantly raising the center of mass. But overall, the robot outperformed expectations.

A Physics Calculations

A.1 Drivetrain

Variables

Variable	Description	Value / Units
m	Robot mass	2.7 kg
g	Gravitational acceleration	9.81 m s ⁻²
θ	Ramp angle	30°
F_{slope}	Force required to climb slope	N
r_{wheel}	Wheel radius	m
T_{total}	Total torque required at wheels	N·m
$T_{gearbox,required}$	Torque required per gearbox	N·m
T_s	Stall torque at 1.2 V	0.370 N·m
T_6	Stall torque at 6 V	≈ 1.85 N·m
$T_{max\ power}$	Torque at maximum power point	N·m
$\omega_{required}$	Required wheel angular velocity	rad/s
v	Target climbing speed	0.4 m s ⁻¹
$\omega_{no\ load}$	No-load speed at 6 V	275 RPM
$\omega_{max\ power}$	Motor speed at maximum power point	RPM
GR	Gear ratio	–
T_{wheel}	Torque delivered at wheel	N·m

Table 6: Variable definitions for drivetrain analysis.

A.1.1 Force Required to Climb Slope

$$F_{slope} = mg \sin \theta = 2.7 \times 9.81 \times \frac{1}{2} = 13.2435 \text{ N} \quad (1)$$

A.1.2 Torque Required at Wheels

The wheel radius is:

$$r_{wheel} = \frac{3.7 \text{ in}}{2} = 1.85 \text{ in} = 0.047 \text{ m} \quad (2)$$

The total torque required across all wheels is:

$$T_{total} = r_{wheel} \times F_{slope} = 13.2435 \times 0.047 = 0.622 \text{ N}\cdot\text{m} \quad (3)$$

Since there are two gearboxes, the torque required per gearbox is:

$$T_{gearbox,required} = \frac{0.622}{2} = 0.311 \text{ N}\cdot\text{m} \quad (\text{per gearbox}) \quad (4)$$

A.1.3 Motor Torque at Maximum Power

The motor has a stall torque of $T_s = 0.370 \text{ N}\cdot\text{m}$ at 1.2 V. Since a 6 V motor is used:

$$T_6 \approx 1.85 \text{ N}\cdot\text{m} \quad (5)$$

Operating at the maximum power point:

$$T_{max \text{ power}} = \frac{T_s}{2} = \frac{1.85}{2} = 0.925 \text{ N}\cdot\text{m} \quad (6)$$

A.1.4 Gear Ratio Selection

The robot is required to climb the ramp at a speed of approximately 0.4 m s^{-1} . The required wheel angular velocity is:

$$\omega_{required} = \frac{v}{r} = \frac{0.4}{0.047} = 8.51 \text{ rad/s} = 81.27 \text{ RPM} \quad (7)$$

The no-load speed at 6 V is 275 RPM, therefore the motor speed at the maximum power point is:

$$\omega_{max \text{ power}} = \frac{275}{2} = 137.5 \text{ RPM} \quad (8)$$

The required gear ratio is:

$$GR = \frac{\omega_{required}}{\omega_{motor}} = \frac{81.27}{137.5} = 0.59 \implies \text{Use } \frac{2}{3} \text{ gear ratio} \quad (9)$$

A $\frac{2}{3}$ gear ratio is selected as this is the closest available ratio.

A.1.5 Verification with $\frac{2}{3}$ Gear Ratio

The torque delivered at the wheel with the chosen gear ratio is:

$$T_{wheel} = 0.925 \times \frac{3}{2} = 1.3875 \text{ N}\cdot\text{m} > 0.311 \text{ N}\cdot\text{m} \checkmark \quad (10)$$

A.2 Scoop Mechanism

Variables Variables

Symbol	Description	Value / Units
T_s	Stall torque	1.85 N·m
ω_0	No-load speed	275 RPM
R	Distance to centre of scoop	2.5 in = 0.0635 m
m	Mass of object	0.2 kg
g	Gravitational constant	9.81 m s ⁻²
$T_{required}$	Torque required to lift object	N·m
$\omega_{required}$	Maximum required scoop speed	70 RPM
GR	Gear ratio	–
T_{scoop}	Torque available at scoop output	N·m

Table 7: Variable definitions for scoop mechanism analysis.

A.2.1 Torque Required to Lift Object

$$T_{required} = mg \times R = 0.2 \times 9.81 \times 0.0635 = 0.125 \text{ N}\cdot\text{m} \quad (11)$$

A.2.2 Gear Ratio Selection

The maximum output speed requirement gives the gear ratio directly:

$$GR = \frac{\omega_{required}}{\omega_0} = \frac{70}{275} = 0.2545 \approx \frac{1}{4} \quad (12)$$

Therefore a $\frac{1}{4}$ gear ratio is selected.

A.2.3 Torque Available at Scoop Output

With a $\frac{1}{4}$ gear ratio:

$$T_{scoop} = T_{max\ power} \times 4 = \frac{T_s}{2} \times 4 = 1.85 \times 2 = 3.7 \text{ N}\cdot\text{m} \quad (13)$$

$$3.7 \text{ N}\cdot\text{m} > 0.125 \text{ N}\cdot\text{m} \checkmark \quad (14)$$

A.3 Shaft Stress Analysis

Shaft used: $\frac{1}{4}$ " aluminium 6061-T6 hex shaft

Variables

Symbol	Description	Value / Units
σ_y	Yield strength	207 MPa
σ_u	Ultimate strength	290 MPa
T	Maximum torque applied	0.125 N·m
c_{max}	Maximum distance from centre	3.667 mm = 3.667×10^{-3} m
a	Hexagon side length	mm
$A_{hexagon}$	Cross-sectional area of hexagon	m ²
J	Polar moment of inertia	m ⁴
$\tau_{torsion}$	Torsional shear stress	MPa
τ_{shear}	Direct shear stress	MPa
τ_{total}	Total shear stress	MPa
F_{shear}	Transverse shear force	N

Table 8: Variable definitions for shaft strength analysis.

A.3.1 Hexagonal Cross-Section Geometry

For a regular hexagon with across-flats distance of 0.25 in = 6.35 mm:

$$a = \frac{6.35}{\sqrt{3}} = 3.667 \text{ mm} \quad (15)$$

$$A_{hexagon} = \frac{3\sqrt{3}}{2} a^2 = \frac{3\sqrt{3} \times 3.667^2}{2} = 34.97 \text{ mm}^2 = 3.497 \times 10^{-5} \text{ m}^2 \quad (16)$$

A.3.2 Moment of Inertia

$$J = \frac{5\sqrt{3}}{16} a^4 = \frac{5\sqrt{3}}{16} (3.667)^4 = 97.67 \text{ mm}^4 = 9.767 \times 10^{-11} \text{ m}^4 \quad (17)$$

A.3.3 Torsional Shear Stress

$$\tau_{torsion} = \frac{T \cdot c_{max}}{J} = \frac{0.125 \times 3.667 \times 10^{-3}}{9.767 \times 10^{-11}} = 4.69 \text{ MPa} \quad (18)$$

A.3.4 Shear Stress

$$\tau_{shear} = \frac{F_{shear}}{A} = \frac{0.2 \times 9.81}{3.497 \times 10^{-5}} = 0.056 \text{ MPa} \quad (19)$$

A.3.5 Total Shear Stress

$$\tau_{total} = \tau_{torsion} + \tau_{shear} = 4.746 \text{ MPa} \quad (20)$$

A.3.6 Factor of Safety

$$\text{Factor of Safety} = \frac{\tau_{yield}}{\tau_{total}} = \frac{207}{4.746} = \boxed{43.62} \quad (21)$$

B Diagrams and Supporting Graphics

B.1 Free Body Daigrams

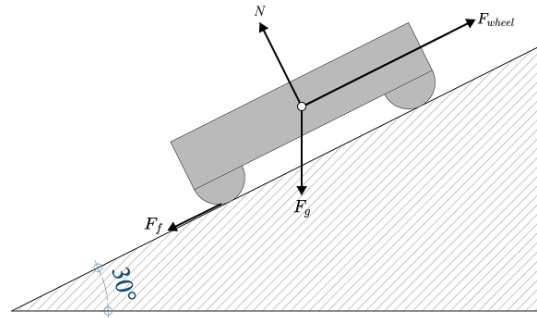
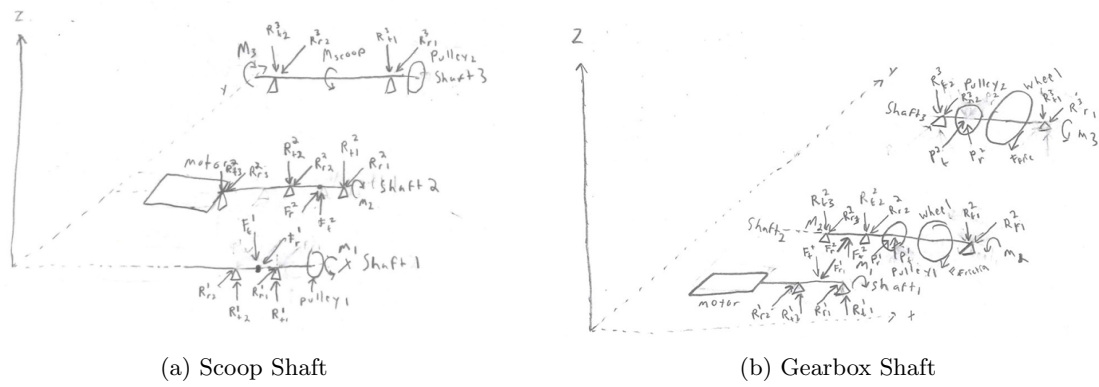


Figure 6: Stability Free Body Diagram



(a) Scoop Shaft

(b) Gearbox Shaft

Figure 7: Shaft Forces Diagram

Item	Part Number	Material	Manufacturing	Qty
11	Belt Pulley	PLA	3D Printing	4
12	Belt Tensioner Right	PLA	3D Printing	2
13	Plastic Sleeve Bearing	Nylon	Parts Bin	11
14	Drivetrain Gearbox	PLA	3D Printing	2
15	1.00in Threaded Standoff	6061 Aluminum	Parts Bin	10
16	0.0625in Hex Spacer	PLA	3D Printing	6
17	24T Gear Spacer (0.125)	Delrin	Parts Bin	2
18	Inch - Spur gear 32DP 58T 14.5PA 0.25in	Delrin	Parts Bin	4
19	0.31250 Bearing	Delrin	Lathe	2
20	Motor Input Shaft	6061 Aluminum	Lathe	2
21	Motor Key Shaft (0.25in Cut)	Steel	Electric Screwdriver	2
22	Back Drive Shaft	6061 Aluminum	Lathe	2
23	91772A118	6061 Aluminum	Parts Bin	16
24	90631A005	6061 Aluminum	Parts Bin	44
25	0.75in Spacer	Nylon	Parts Bin	12
26	Drivetrain Belt	Rubber	Parts Bin	2
27	Scoop Belt	Rubber	Parts Bin	1
28	91772A112	6061 Aluminum	Parts Bin	10
29	4-40 Washer	6061 Aluminum	Parts Bin	16
30	Ramp Support Bracket (Left)	PLA	3D Printing	1
31	Scoop Gearbox	PLA	3D Printing	1
32	Scoop Motor Mount	PLA	3D Printing	1

Item	Part Number	Material	Manufacturing	Qty
33	Intake Gearbox Shaft Support	PLA	3D Printing	1
34	Scoop Side Support	Acrylic	Laser Cut	2
35	Belt Pulley Scoop	PLA	3D Printing	1
36	Sweeper Receiver	PLA	3D Printing	1
37	Scoop	PLA	3D Printing	1
38	Scoop Shaft	6061 Aluminum	Lathe	1

Table 9: Bill of Materials

D Technical Drawings

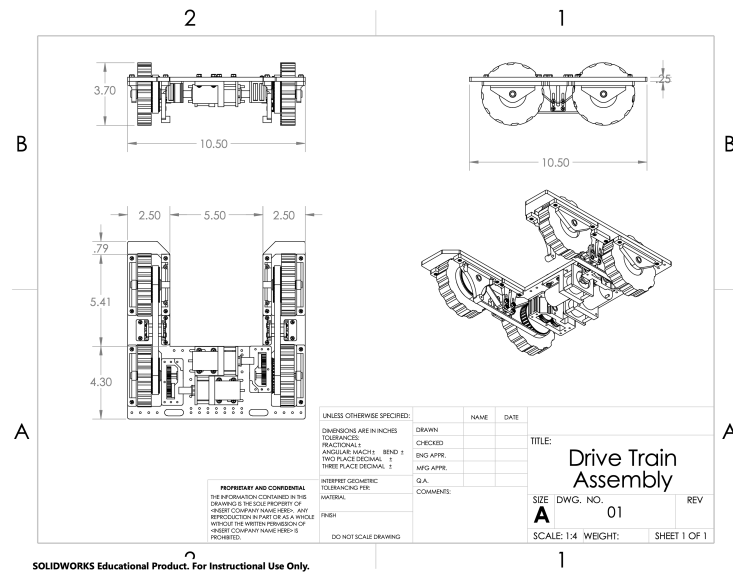


Figure 9: Drive Train., Assembly Drawing

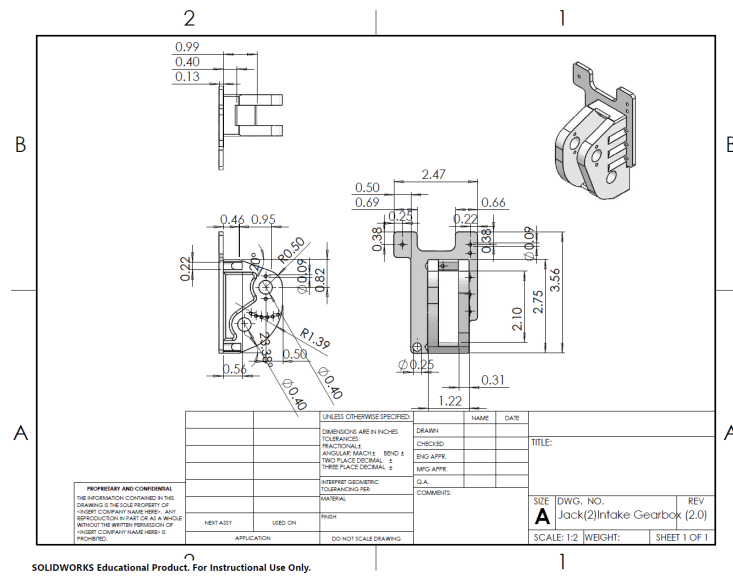


Figure 10: Scoop Gearbox Support Piece., Assembly Drawing

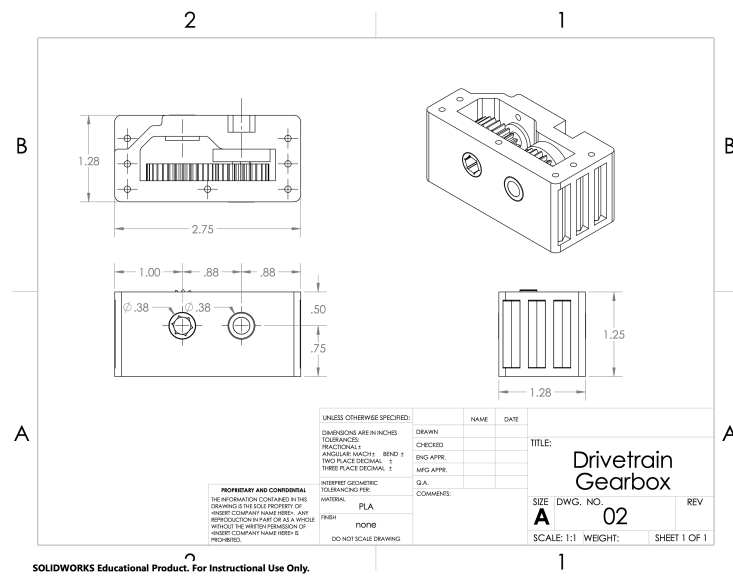


Figure 11: Drive Train Gear Box., Assembly Drawing

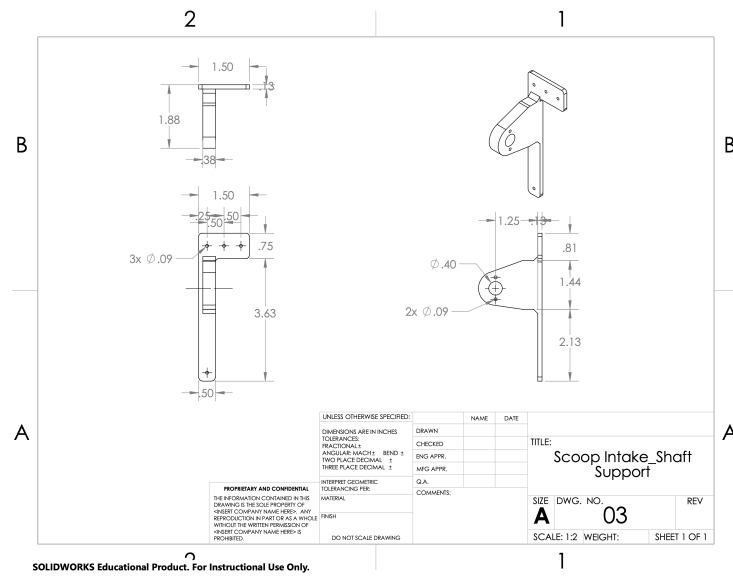


Figure 12: Shaft Support., Part Drawing

E Assembly Drawing

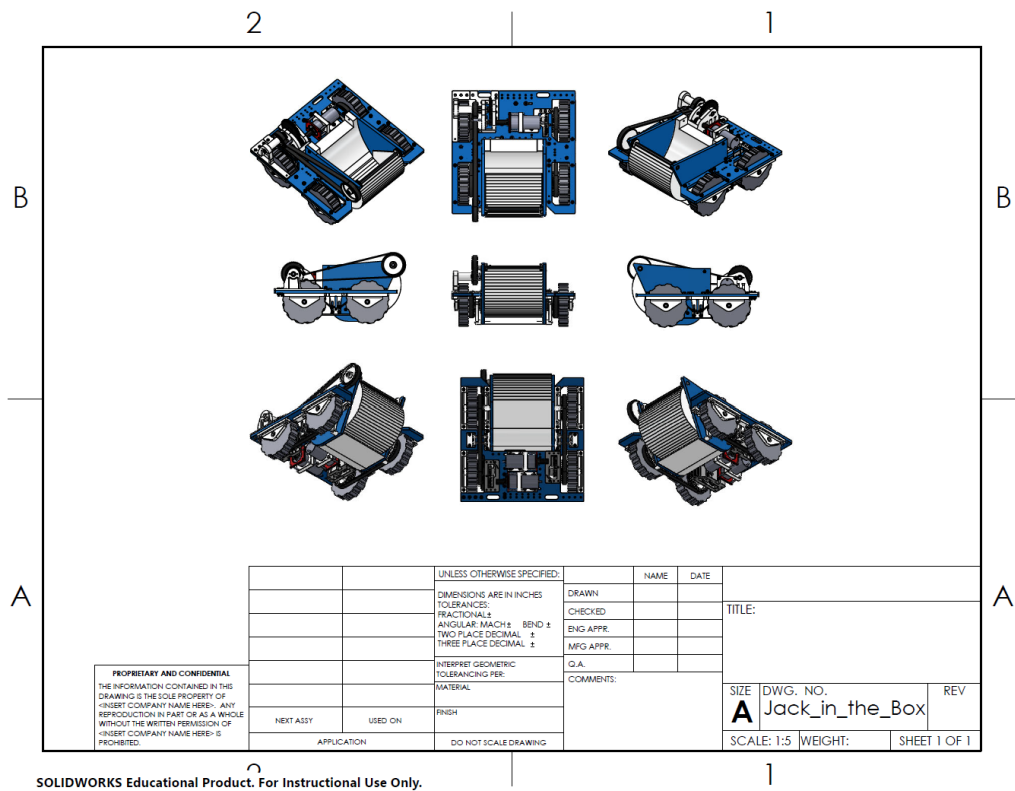


Figure 13: Full assembly drawing of the Sooper Scooper 5000.

F Exploded View

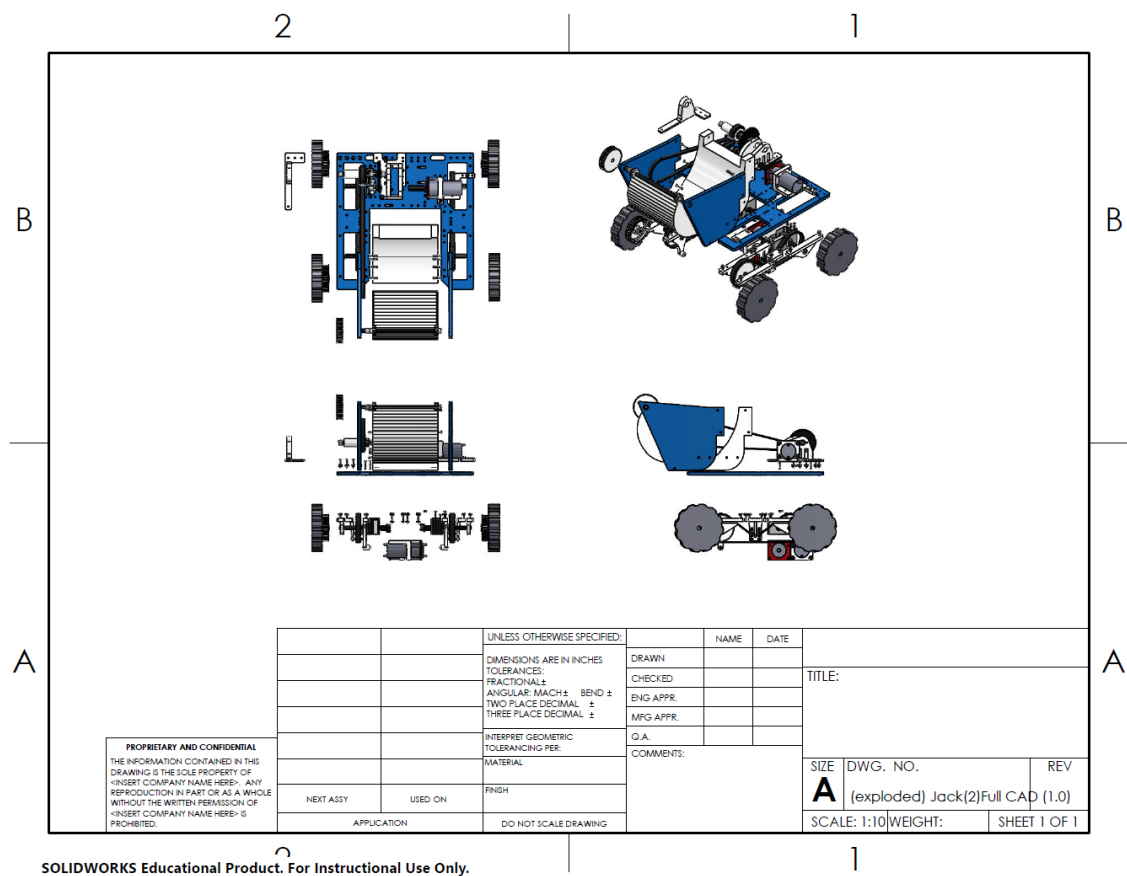
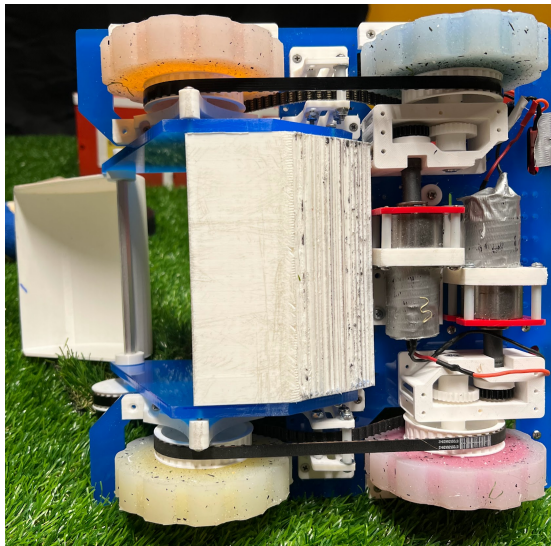
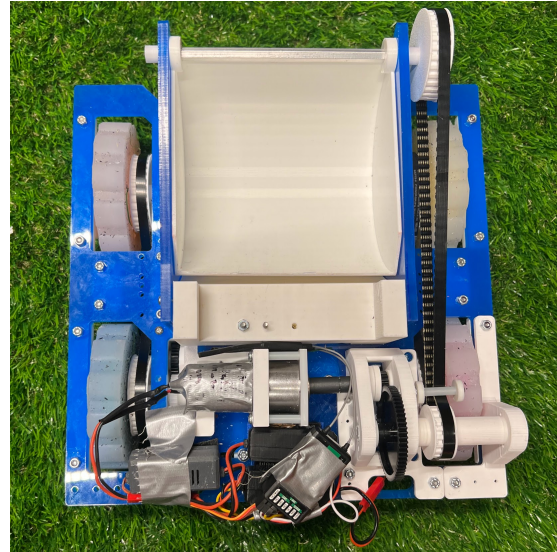


Figure 14: Exploded view of robot assembly

G Final Photos



(a) Final Bottom View



(b) Final Top View

Figure 15: Final Robot Views

H Contributions

Member	Contributions
Charlie	Physics calculations, design iterations, bill of materials, free body diagrams
Gabriela	Abstract, introduction, concept development, final system performance analysis, technical drawings and supporting diagrams
Ian	SolidWorks CAD modeling, exploded view, and assembly drawing
Alex	Photography, design iterations and prototyping, Pugh matrix
Daniel	Shaft force analysis, free body diagrams
